

10 (short)

A company supplying mineral water is considering switching from PET polymer bottles to aluminium (Al) alloy cans. A life cycle analysis shows that the embodied energy of the container material is the most significant component of the total energy usage, per unit volume of water contained. Data for each container are given in Table 1.

- (a) Calculate the embodied energy of each container, per unit volume of water contained, if they are manufactured from 100% virgin material. [2]
- (b) Calculate the fraction of recycled Al alloy that must be used to manufacture each can for it to have the same embodied energy as a PET bottle, per unit volume of water contained, if the PET bottles use 30% recycled material, a typical maximum. [4]
- (c) Outline two strategies to reduce the embodied energy of the Al alloy cans, per unit volume of water contained. In your answer, explain briefly the technical challenges the strategy presents. [4]

	PET bottle	Al alloy can
Container capacity ($\text{m}^3 \times 10^{-6}$)	500	330
Container mass ($\text{kg} \times 10^{-3}$)	30	12
Embodied energy, virgin material (MJ kg^{-1})	84	200
Embodied energy, recycled material (MJ kg^{-1})	39	25

Table 1